

WEEKLY ASSESSMENT-1 - Answers

1. Difference between List and Tuple

List: Mutable, uses [] and more memory. Tuple: Immutable, uses () and less memory.

2. Swap Two Variables

Python allows swapping in one line: a, b = b, a

3. Data Type of Elements in a Mixed List

Use a loop and type() function to print the data type of each element.

4. Palindrome Program

Check whether a string is equal to its reverse using text == text[::-1].

5. Difference between / and //

/ gives normal division (float). // gives floor division (integer quotient).

6. Last 3 Characters of a String

Use slicing: text[-3:]

7. Count Vowels

Loop through the sentence and count characters in 'aeiou'.

8. f-Strings

f-strings are used to insert variables directly inside strings, e.g., f'My name is {name}'

9. Print Even Numbers from 1 to 20

Create a list using range(1,21) and print numbers divisible by 2.

10. Difference between range(5) and list(range(5))

range(5) returns a range object; list(range(5)) returns [0,1,2,3,4].